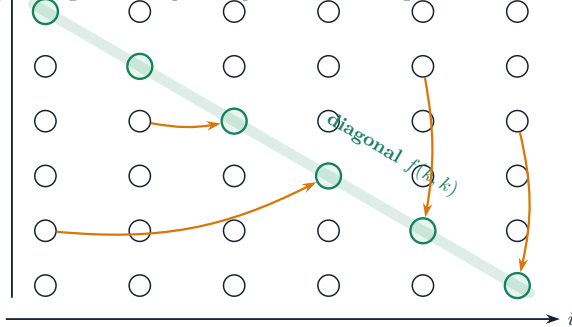


A whole grid can collapse to its diagonal

The indexed family as a grid

every off-diagonal value points upward to some diagonal value



COFINAL

$$\forall i, j, \exists k, f(i, j) \leq f(k, k)$$

ENOUGH

the diagonal already reaches every height

Before: two independent choices

$$\bigvee_i \bigvee_j f(i, j)$$

scan every row, then join the row joins



After: one synchronized choice

$$\bigvee_k f(k, k)$$

walk only along the diagonal

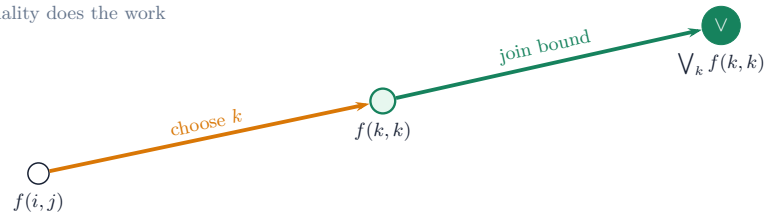
$$\left(\bigvee_i \bigvee_j f(i, j) \right) = \bigvee_k f(k, k)$$

iSup2_eq_iSup_diagonal

Why the equality is true: two inequalities

HARD DIRECTION $\bigvee_i \bigvee_j f(i, j) \leq \bigvee_k f(k, k)$

cofinality does the work

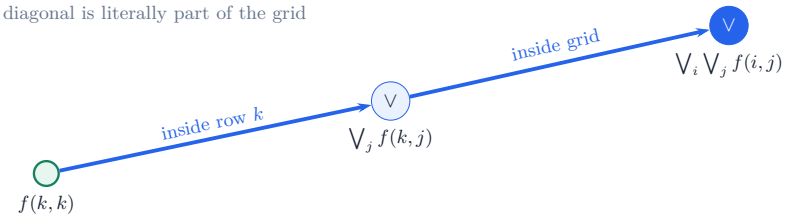


Every $f(i, j)$ is below the diagonal join.

therefore the join of all $f(i, j)$ is below it too

EASY DIRECTION $\bigvee_k f(k, k) \leq \bigvee_i \bigvee_j f(i, j)$

the diagonal is literally part of the grid



Every diagonal term occurs at grid position (k, k) .

join the pointwise inequalities over k

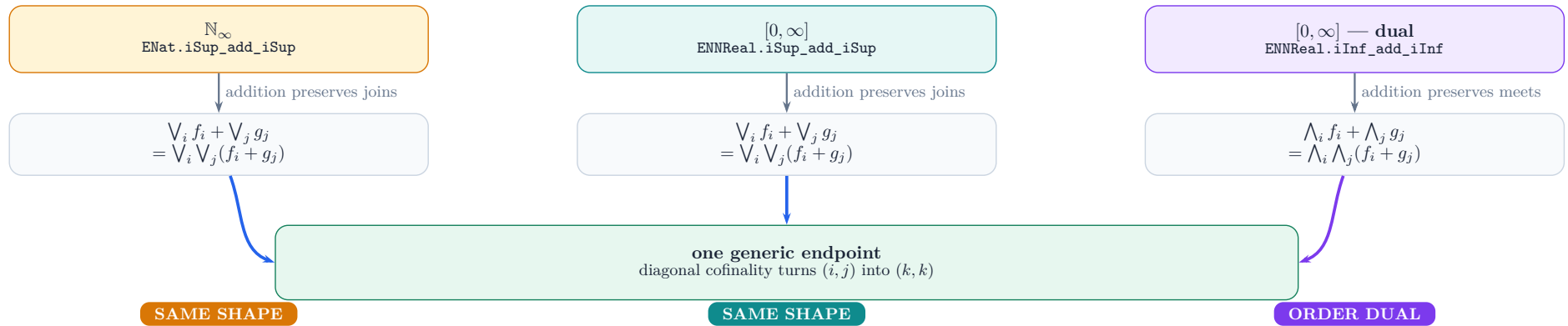
BIG IDEA

Cofinality proves one side; inclusion of the diagonal proves the other.

Three duplicated proofs become one lattice move

PR 1 COMPLETE LATTICES

The common pipeline



BEFORE

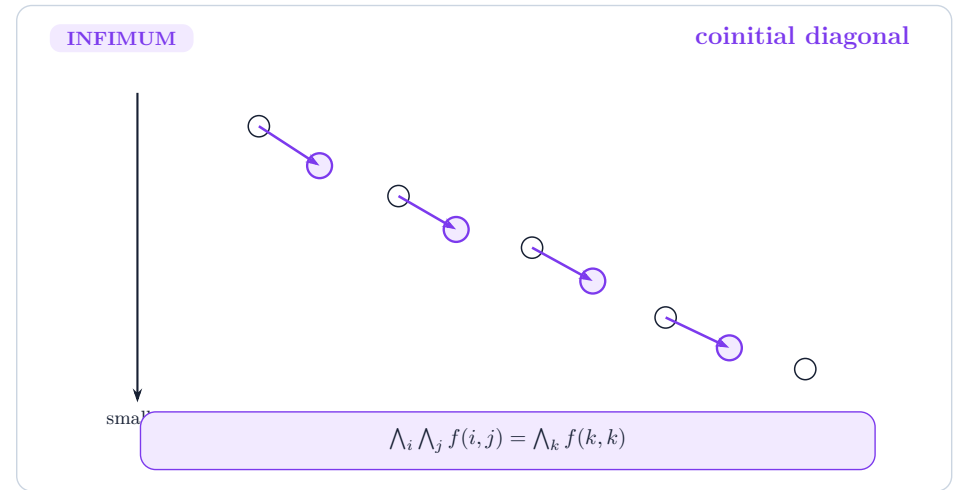
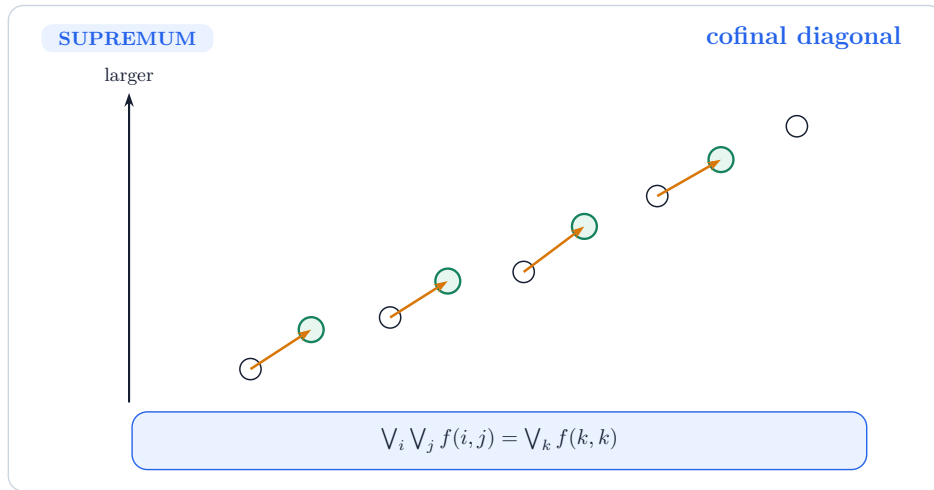
repeat nested-supremum bounds and witness chasing inside each numeric namespace

AFTER

`exact iSup2_eq_iSup_diagonal ... h`
after the domain-specific distribution step

The order-dual mirror gives the infimum theorem

PR 1 COMPLETE LATTICES

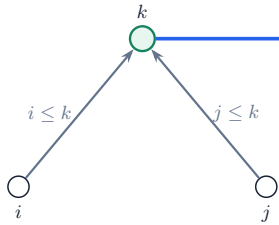


When does the diagonal condition appear naturally?

PR 1 COMPLETE LATTICES

Directed indices synchronize independent choices

choose one later index k that dominates both i and j



monotone families
 $f_i \leq f_k, \quad g_j \leq g_k$

monotonicity of addition
 $f_i + g_j \leq f_k + g_k$

Two independent approximations can be synchronized at one later stage.

Cool facts

NO CHAIN NEEDED

The lattice need not be linearly ordered.
Only joins and the witness k matter.

NO ARITHMETIC INSIDE

Addition is used before the generic lemma.
The collapse itself is pure order theory.

EMPTY INDEX IS SAFE

In a complete lattice, both empty joins
are bottom, so the equality still holds.

WITNESSES MAY VARY

k may depend on both i and j ;
no uniform choice is required.

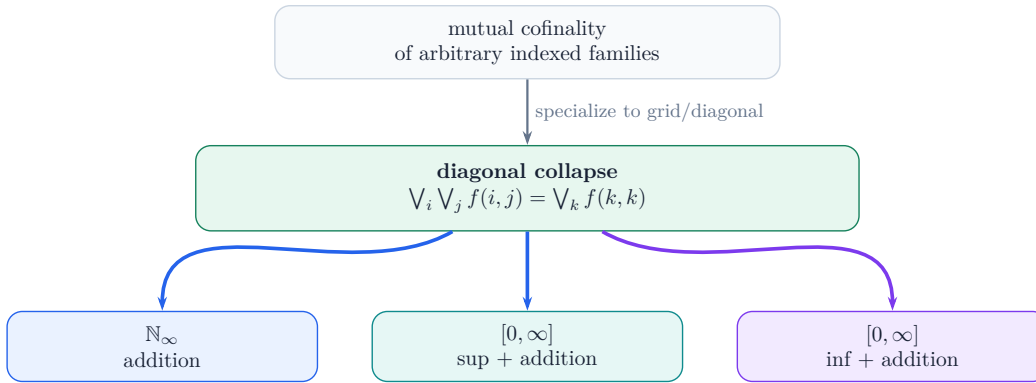
BIG IDEA

The lemma packages the synchronization step, not the domain-specific algebra.

Why this is an upstream-sized contribution

PR 1 COMPLETE LATTICES

A small API ladder



WEAK

only a complete join-semilattice

no ENat, ENNReal, addition, or linear order

NAMED

predictable indexed-supremum API

iSup2_eq_iSup_diagonal

REUSED

three existing proofs simplify

two join proofs + their meet-side dual

SPOT THE GRID

ERASE ARITHMETIC

NAME DIAGONAL COLLAPSE

REFACTOR 3 PROOFS